

University of Heidelberg
Faculty of Physics and Astronomy
Beginners Physics Lab 1

Experiment 15 – Inclined Plane

Author:	Jonathan Gießen
Group:	Group 8
Supervisor:	Matha Reece
Experiment Date:	15. September 2026



Contents

1. Introduction	3
2. Basic Principles and theories	3
2.1. Dynamics and Acceleration	3
2.2. Conservation of Energy	3
2.3. Gaussian error propagation	4
3. Measuring Protocol	5
3.1. Setup and Tasks	5
3.2. Execution of the Experiment	9
4. Evaluation	9
4.1. Exercise 1: Comparison of Acceleration (theoretical vs. experimental) . .	9
4.2. Exercise 2: Calculating the kinetic Energy	12
5. Discussion of Results	13
6. Conclusion	14
A. Code	15

AI-Disclaimer

AI tools were used to assist in translating, as well as in the generation of text in section 1

1. Introduction

The experiment investigates the fundamental mechanics of rigid bodies moving down an inclined plane. By observing different rolling bodies specifically a solid cylinder and a hollow cylinder the experiment explores how mass distribution affects rotational and translational motion. The main objectives are to quantitatively determine the acceleration of these bodies as they roll down the ramp and to verify the law of conservation of mechanical energy by comparing the initial potential energy with the final translational and rotational kinetic energy.

2. Basic Principles and theories

The motion of a round body rolling down an inclined plane without slipping is governed by both linear and rotational dynamics. Two primary physical frameworks are utilized to analyze this motion: Newton's laws of motion and the principle of energy conservation.

2.1. Dynamics and Acceleration

When a body rolls down an inclined plane with an angle φ , it experiences a downhill force component from gravity ($F_H = mg \sin \varphi$) and an opposing static friction force (F_R). This friction force exerts a torque (moment of force) M relative to the center of mass, causing the body to rotate. The relationship between torque, the moment of inertia I , and angular acceleration $\frac{d\omega}{dt}$ is given by:

$$M = F_R \cdot r = I \frac{d\omega}{dt} \quad (2.1)$$

Using the rolling without slipping condition, the velocity of the center of gravity is $v_s = r\omega$. Differentiating this yields the relationship between linear acceleration a_s and angular acceleration, allowing us to express the friction force as:

$$F_R = \frac{I}{r^2} \cdot a_s \quad (2.2)$$

Applying Newton's second law for linear motion along the plane ($ma_s = mg \sin \varphi - F_R$) and substituting the friction force, the linear acceleration of the rolling body is derived as:

$$a_s = \frac{mg \sin \varphi}{m + \frac{I}{r^2}} \quad (2.3)$$

This formula demonstrates that the acceleration depends on the body's moment of inertia, explaining why a hollow cylinder and a solid cylinder roll at different rates despite having the same radius.^[1]

2.2. Conservation of Energy

The motion can also be analyzed through the conservation of mechanical energy. Assuming no energy is lost to non-conservative forces (like air resistance or kinetic friction, since the static friction does no work), the initial potential energy is fully converted into kinetic

energy. The total energy W_{ges} is the sum of translational kinetic energy, rotational kinetic energy, and potential energy:

$$W_{ges} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + mgh \quad (2.4)$$

As the body descends a vertical height h , the decrease in potential energy (mgh) equals the increase in total kinetic energy ($\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$).

2.3. Gaussian error propagation

The errors of the measured quantities are propagated using the Gaussian error propagation formula:

$$\sigma_f = \sqrt{\sum_{i=1}^N \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2} \quad (2.5)$$

as well as the addition of statistical and systematic errors in quadrature:

$$\sigma_{total} = \sqrt{\sigma_{stat}^2 + \sigma_{sys}^2} \quad (2.6)$$

3. Measuring Protocol

3.1. Setup and Tasks

Protokoll: Experiment 15: Inclined Plane

Jonathan Gießen, Theo Senoner

15.9.2026

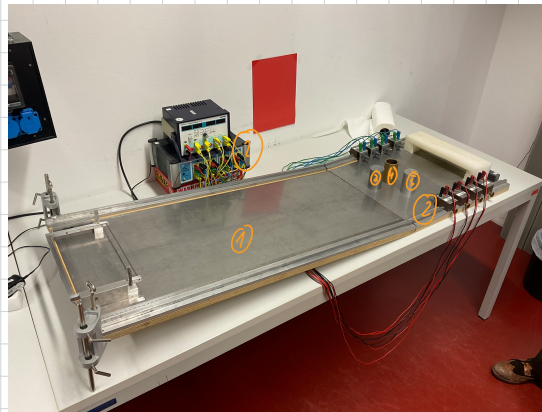


Bild 3.1 Setup

Setup :

- Height-adjustable runway ①
- solid cylinder: alu , $\rho = 2,70 \frac{\text{g}}{\text{cm}^3}$ ③
- caliper
- Light-barrier with stopwatch ②
- hollow cylinder: brass , $\rho = 8,44 \frac{\text{g}}{\text{cm}^3}$ ④
- Scale
- Spirit level
- composite cylinder: sheath of aluminum, core of brass ⑤

A1: Measurement

Tab 3.1: Measurement of rolling bodies

cylinder	d _{inner} [mm]	d _{outer} [mm]	mass [g]
solid	0	$5,00 \pm 0,05$	$443,0 \pm 1,0$
hollow	$4,65 \pm 0,05$	$5,00 \pm 0,05$	$443,0 \pm 1,0$
(composite)	$0,76 \pm 0,05$	$5,00 \pm 0,05$	$443,0 \pm 1,0$

length of track [cm]: $100,0 \pm 0,5$

With the error estimation based on the error of the ruler and the uncertainty of the measurement method (the plane the ruler is set on is not perfectly flat and there is a tiny gap between the measured length and the actual plane crossed)

$$\Delta \sigma = \frac{\sigma}{\sqrt{n}} \quad \Delta C = 0,05 \text{ cm}$$

We adjust for the error of the caliper not being upright by doing multiple measurements.

$$\text{Total error: } \Delta = \sqrt{\Delta \sigma^2 + \Delta C^2}$$

Height of track [cm]:

1: 11,10

2: 12,80

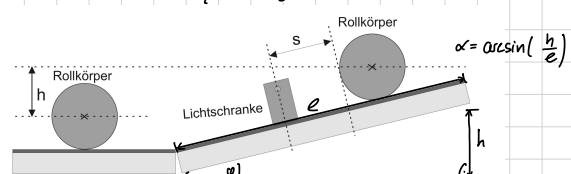
3: 11,50

4: 12,80

5: 12,70

$$\bar{h} = (12,2 \pm 0,4) \text{ cm}$$

with $\Delta \sigma = 0,37$



The time error for all time measurements is $\Delta t = \pm 0,001s$

A2: movement type? consistent acceleration

$$L_i = L_{\text{end}} - L_{\text{start}}, \Delta r_{\text{uler}} = 0,05 \text{ cm}$$

Tbl 3.2 Full cylinder

barrier [cm]	Δr_{tot}	time [s]
$L_1 = 25,00 \pm 0,07$		$t_1 = 0,821$
$L_2 = 36,00 \pm 0,07$		$t_2 = 1,084$
$L_3 = 49,00 \pm 0,07$		$t_3 = 1,272$
$L_4 = 64,00 \pm 0,07$		$t_4 = 1,451$

Hollow cylinder

barrier [cm]	Δr_{tot}	time [s]
$L_1 = 25,00 \pm 0,07$		$t_1 = 1,011$
$L_2 = 36,00 \pm 0,07$		$t_2 = 1,204$
$L_3 = 49,00 \pm 0,07$		$t_3 = 1,403$
$L_4 = 64,00 \pm 0,07$		$t_4 = 1,604$

The uncertainty of the light source not being perfectly centered is neglected in the face of the much bigger uncertainty of the wler. Nevertheless it is worth for a mention.

We aligned both sides (sender and receiver) with the wler so no alignment error aside from Δr is expected.

We add the two separate Δr :

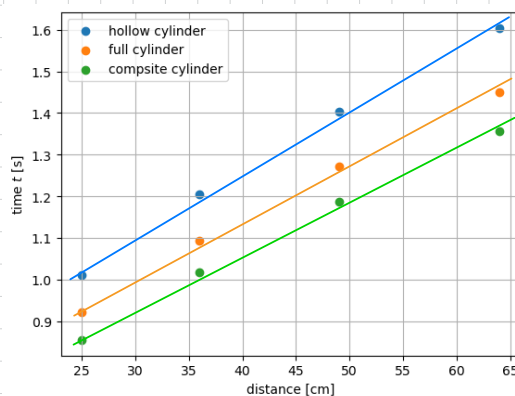
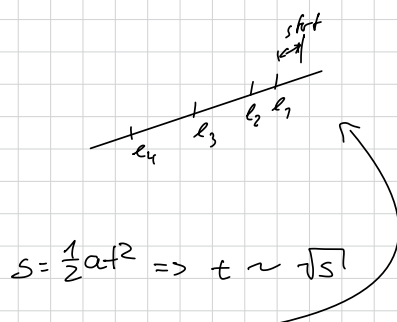
$$\Delta L_{\text{tot}} = \sqrt{\Delta r_1^2 + \Delta r_2^2} = \sqrt{2} \cdot 0,05 =$$

Δr_1 : sender side, Δr_2 : receiver side

Composed cylinder

barrier [cm]	Δr_{tot}	time [s]
$L_1 = 25,00 \pm 0,07$		$t_1 = 0,855$
$L_2 = 36,00 \pm 0,07$		$t_2 = 1,018$
$L_3 = 49,00 \pm 0,07$		$t_3 = 1,186$
$L_4 = 64,00 \pm 0,07$		$t_4 = 1,356$

Plots:



Tab: 3.3 Measurement of acceleration											
A3 [cm]	dist $\Delta t_{\text{tot}} = 0.07$	time hollow $\Delta t =$ [ms]					+ full [ms]				
$l_1 = 25$		1,019	1,027	1,014	1,022	1,009	0,92	0,914	0,900	0,904	0,913
$l_2 = 36$		1,212	1,223	1,209	1,215	1,201	1,096	1,089	1,075	1,074	1,087
$l_3 = 49$		1,410	1,426	1,410	1,415	1,400	1,278	1,268	1,255	1,254	1,266
$l_4 = 64$		1,610	1,629	1,613	1,614	1,600	1,458	1,466	1,435	1,435	1,446

Table 3.4
Mean + standard error:

	Hollow [s]	Full [s]
l_1 :	$1,018 \pm 0,004$	$0,906 \pm 0,006$
l_2 :	$1,212 \pm 0,004$	$1,080 \pm 0,005$
l_3 :	$1,412 \pm 0,004$	$1,260 \pm 0,006$
l_4 :	$1,613 \pm 0,005$	$1,440 \pm 0,006$

A4:

```

: # messung, t1, t2

# hollow
m1h = np.array([1.922, 2.192])
m2h = np.array([1.951, 2.222])
m3h = np.array([1.936, 2.206])
m4h = np.array([1.922, 2.192])
m5h = np.array([1.938, 2.207])

# full
m1f = np.array([1.724, 1.967])
m2f = np.array([1.736, 1.980])
m3f = np.array([1.718, 1.961])
m4f = np.array([1.737, 1.981])
m5f = np.array([1.749, 1.992])

```

Tbl. 3.5 measurement of translational velocity Hollow:

[cm]	Distance	Δt_{tot}	[s] ΔT_1	[s] ΔT_2	[s] ΔT_3	[s] ΔT_4	[s] ΔT_5
$\Delta L =$	$25,00 \pm 0,07$		0,270	0,271	0,270	0,270	0,269

$$\overline{T_H} = (0,2700 \pm 0,0010) \text{ s with } \Delta \overline{T_H} = \sqrt{\left(\frac{\sigma_{T_H}}{\sqrt{5}}\right)^2 + \Delta t^2} \Rightarrow \overline{v_H} = \frac{\Delta L}{\overline{T_H}}$$

Tbl. 3.6 measurement of translational velocity Full:

[cm]	Distance	Δt_{tot}	[s] ΔT_1	[s] ΔT_2	[s] ΔT_3	[s] ΔT_4	[s] ΔT_5
$\Delta L =$	$25,00 \pm 0,07$		0,243	0,244	0,243	0,244	0,243

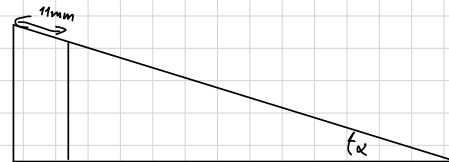
$$\overline{T_F} = (0,2430 \pm 0,0010) \text{ s with } \Delta \overline{T_F} = \sqrt{\left(\frac{\sigma_{T_F}}{\sqrt{5}}\right)^2 + \Delta t^2} \Rightarrow \overline{v_F} = \frac{\Delta L}{\overline{T_F}}$$

Error propagation!



Img 3.2 Distance

Prof. Dr.
16/9/26



$$h = \sin(\alpha) \cdot (l - 1\text{mm})$$

3.2. Execution of the Experiment

The Experimental procedure involves the following main steps:

1. **Setup and Geometry Measurement:** Familiarizing with the light barrier control unit and stopwatch system. Measuring the diameters of the test bodies (solid and hollow cylinders) using a caliper. Using a spirit level to ensure the inclined plane has an even inclination across its entire width, adjusting the supports if necessary (No adjustment was needed in this case).
2. **Qualitative Analysis of Motion:** Verifying uniformly accelerated motion by placing the light barriers at appropriate intervals ($t \propto \sqrt{s}$). Observing and explain qualitatively why bodies with different mass distributions require different amounts of time to reach the bottom.
3. **Quantitative Acceleration Measurement:** Place the light barriers along the incline and measure their exact distances s from the starting mechanism. Release the solid and hollow cylinders and record the time it takes for them to pass the barriers. Repeat this measurement 5 times for each rolling body.
4. **Energy Conservation Study:** To evaluate the conservation of energy, two light barriers are positioned on the horizontal section at the end of the ramp. The final translational velocity of the bodies are measured. The calculated total kinetic energy will then be compared to the initial potential energy, which is derived from the height difference of the plane.

4. Evaluation

4.1. Exercise 1: Comparison of Acceleration (theoretical vs. experimental)

The acceleration of the rolling bodies can be calculated using the derived formula:

$$a_s = \frac{g \sin \varphi}{1 + \frac{I}{mr^2}} \quad (4.1)$$

The Moment of inertia I for a solid cylinder is $I = \frac{1}{2}mr^2$, and for a hollow cylinder, it is $I = \frac{1}{2}m(r_{out}^2 + r_{in}^2)$

With the measured parameters of the cylinders, the theoretical accelerations can be calculated. The parameters are summarized in Table 4.1.

Table 4.1: Parameters of the cylinders used in the experiment.

Parameter	Solid Cylinder	Hollow Cylinder
Mass [g]	443.0 ± 1.0	443.0 ± 1.0
Inner diameter [m]	0	$(46.50 \pm 0.05) \times 10^{-3}$
Outer diameter [m]	$(50.00 \pm 0.05) \times 10^{-3}$	$(50.00 \pm 0.05) \times 10^{-3}$
Moment of Inertia [$\text{kg} \times \text{m}^2$]	$(1.384 \pm 0.004) \times 10^{-4}$	$(2.582 \pm 0.007) \times 10^{-4}$

To calculate the theoretical acceleration, the angle of inclination φ of the ramp is needed. This can be determined from the height difference h and the length of the ramp L using the relation:

$$\sin \varphi = \frac{h}{L} \quad (4.2)$$

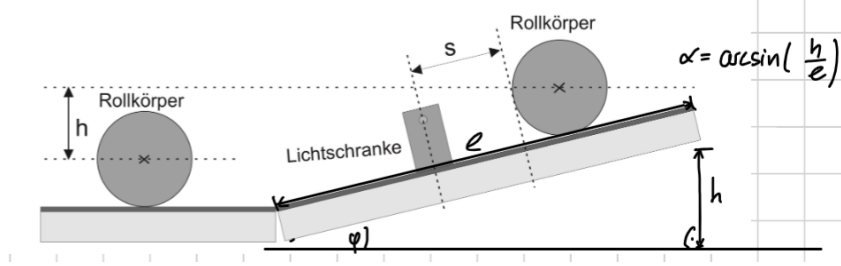


Figure 4.1: Schematic representation of the inclined plane with height h and length L . [1]

with the measured height $h = (10.1 \pm 0.4) \times 10^{-2} \text{ m}^1$ and the length of the track $l = (1.000 \pm 0.005) \text{ m}$, the angle of inclination can be calculated as:

$$\varphi = \arcsin\left(\frac{h}{L}\right) = \arcsin\left(\frac{(10.1 \pm 0.4) \times 10^{-2}}{1.000 \pm 0.005}\right) \quad (4.3)$$

$$\approx 5.797 \pm 0.024^\circ = (0.1012 \pm 0.0004) \text{ rad} \quad (4.4)$$

Therefore, the theoretical accelerations for the solid and hollow cylinders can be calculated as (Equation 2.3):

$$a_{\text{solid}} = \frac{g \sin \varphi}{1 + \frac{I_{\text{solid}}}{mr^2}} = (0.661 \pm 0.026) \frac{\text{m}}{\text{s}^2} \quad (4.5)$$

$$a_{\text{hollow}} = \frac{g \sin \varphi}{1 + \frac{I_{\text{hollow}}}{mr^2}} = (0.513 \pm 0.021) \frac{\text{m}}{\text{s}^2} \quad (4.6)$$

Now, the experimentally determined times are plotted quadratically t^2 against the distance s . The slope of the linear fit corresponds to the acceleration, which can be compared to the theoretical values.

¹The height was measured two days later, because the initial measurement was not accurate.

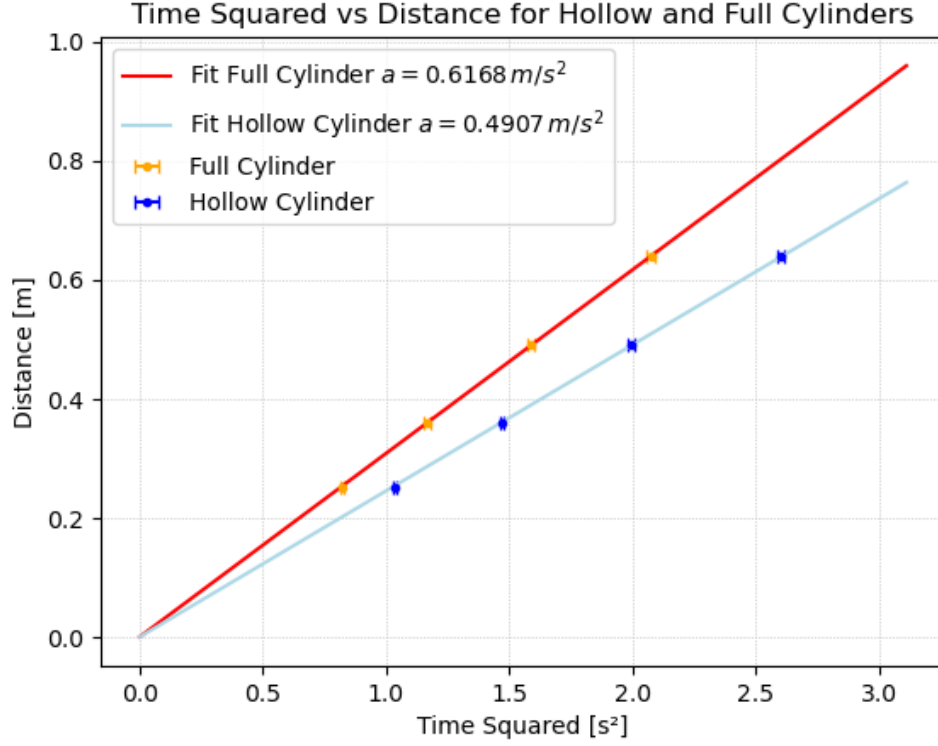


Figure 4.2: Comparison of theoretical and experimental accelerations for solid and hollow cylinders.

The function used for the linear fit is $t^2 = \frac{2}{a}s$, where a is the acceleration. The slope of the fit line gives the value of $\frac{2}{a}$. The accelerations obtained from the fit are:

$$a_{\text{solid, exp}} = (0.617 \pm 0.004) \frac{\text{m}}{\text{s}^2} \quad (4.7)$$

$$a_{\text{hollow, exp}} = (0.4907 \pm 0.0027) \frac{\text{m}}{\text{s}^2} \quad (4.8)$$

The experimental results show that the solid cylinder accelerates faster than the hollow cylinder, which is consistent with the theoretical predictions.

Interestingly, the experimental acceleration of both cylinders is much lower than the theoretical values.

The sigma-deviation of the experimental values from the theoretical ones can be calculated as:

$$\sigma_{\text{solid}} = \frac{|a_{\text{solid, exp}} - a_{\text{solid, theo}}|}{\sigma_{\text{hollow, theo}}} \Rightarrow 1.73\sigma \quad (4.9)$$

$$\sigma_{\text{hollow}} = \frac{|a_{\text{hollow, exp}} - a_{\text{hollow, theo}}|}{\sigma_{\text{solid, theo}}} \Rightarrow 1.19\sigma \quad (4.10)$$

Conclusion

The experimental results for the accelerations of the solid and hollow cylinders show a significant deviation from the theoretical predictions, with both results being lower than expected. The calculated values are:

Value	Solid Cylinder	Hollow Cylinder
Theoretical Acceleration:	0.661 ± 0.026	0.513 ± 0.021
Experimental Acceleration:	0.617 ± 0.004	0.4907 ± 0.0027
Deviation:	1.69σ	1.06σ

4.2. Exercise 2: Calculating the kinetic Energy

For calculating the kinetic energy, the final velocities of the cylinders are measured using two light barriers at the end of the ramp. The time taken to pass between the two barriers is recorded, and the velocity is calculated as:

$$v = \frac{d}{t} \quad (4.11)$$

Through the conservation of energy, the initial potential energy $E_{pot} = mgh$ should equal the total kinetic energy $E_{kin} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$. The rotational kinetic energy can be expressed in terms of the linear velocity using the rolling condition $\omega = \frac{v}{r}$.

The height difference equation can be derived from the geometry and the starting distance from the end of the ramp.



Figure 4.3: Schematic representation of the height difference h and the length of the ramp L .

For the height difference h , we can use the following relation (intercept theorem):

$$h_{eff} = \frac{h}{l} \cdot (l - 11\text{mm}) = (0.100 \pm 0.004) \text{ m} \quad (4.12)$$

The potential energy for the cylinders equals:

$$E_{pot} = (0.443 \pm 0.001) \text{ kg} \cdot 9.81 \frac{\text{m}}{\text{s}^2} \cdot (0.100 \pm 0.004) \text{ m} = (0.434 \pm 0.018) \text{ J} \quad (4.13)$$

The measured times and speeds for the solid and hollow cylinders with $r = (25.000 \pm 0.025) \times 10^{-3} \text{ m}$ to pass between the two light barriers with a distance of $d = (25.00 \pm 0.07) \times 10^{-2} \text{ m}$ are:

Table 4.2: Measured times and calculated speeds for the solid and hollow cylinders.

Cylinder Type	Time [s]	Speed [m/s]	ω [rad/s]
Solid Cylinder	0.2430 ± 0.0010	1.0271 ± 0.0051	41.08 ± 0.21
Hollow Cylinder	0.2700 ± 0.0010	0.9259 ± 0.0043	37.04 ± 0.18

The kinetic energies for the solid and hollow cylinders can be calculated as follows:

$$E_{kin} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \quad (4.14)$$

This results in the following kinetic energies for the solid and hollow cylinders:

Table 4.3: Calculated kinetic energies for the solid and hollow cylinders.

Cylinder Type	Translational KE [J]	Rotational KE [J]	Total KE [J]
Solid Cylinder	0.2337 ± 0.0024	0.1168 ± 0.0012	0.351 ± 0.004
Hollow Cylinder	0.1900 ± 0.0019	0.1771 ± 0.0018	0.367 ± 0.004

Comparing the total kinetic energies with the initial potential energy, we get:

$$E_{kin,solid} = (0.351 \pm 0.004) \text{ J} \quad (4.15)$$

$$E_{kin,hollow} = (0.367 \pm 0.004) \text{ J} \quad (4.16)$$

$$E_{pot} = (0.434 \pm 0.018) \text{ J} \quad (4.17)$$

This results in a deviation of:

$$\Delta E_{solid} = E_{pot} - E_{kin,solid} = (0.083 \pm 0.018) \text{ J} \quad (4.18)$$

$$\Rightarrow 4.61\sigma \quad (4.19)$$

$$\Delta E_{hollow} = E_{pot} - E_{kin,hollow} = (0.067 \pm 0.018) \text{ J} \quad (4.20)$$

$$\Rightarrow 3.72\sigma \quad (4.21)$$

Conclusion

The total kinetic energies for both cylinders are lower than the initial potential energy, indicating that some energy was lost during the rolling motion:

Value	Solid Cylinder	Hollow Cylinder
Initial Potential Energy:	0.434 ± 0.018	0.434 ± 0.018
Total Kinetic Energy:	0.351 ± 0.004	0.367 ± 0.004
Deviation:	4.61σ	3.72σ

5. Discussion of Results

In figure 4.2 you can see that the acceleration of the solid cylinder is higher than that of the hollow cylinder, which is consistent with the theoretical predictions. This is due to

the fact that the solid cylinder has a lower moment of inertia, allowing it to accelerate more quickly down the incline.

Even though the experimental results show the expected trend, there are significant deviations from the theoretical values. The experimental accelerations for both cylinders are lower than the theoretical predictions. The kinetic energies show deviations from the theoretical predictions as well. The lower experimental accelerations suggest that additional resistive forces, such as rolling friction may have influenced the motion of the cylinders. The total kinetic energies for both cylinders are lower than the initial potential energy, indicating that some energy was lost during the rolling motion, likely due to friction.

Since friction is proportional to the normal force, which is in turn proportional to the mass of the rolling body, it is expected that the solid cylinder would experience a greater frictional force than the hollow cylinder. This could explain why the solid cylinder has a lower kinetic energy than the hollow cylinder, despite having a higher acceleration, as well as bigger deviations from the theoretical predictions, which neglect resistive forces. The solid cylinder's greater mass results in a larger normal force, leading to increased frictional losses during its motion down the incline.

It is very unlikely that the timing and measurement errors are the main contributors to the observed deviations, as the experimental setup was carefully calibrated and repeated measurements were taken to minimize random errors. I suggest that the measurement of the height difference should be repeated, as it was measured two days later and could have been affected by systematic errors, because the height difference that, was measured in the beginning was a lot higher than the one measured two days later. Identifying and measuring the height of the slope was not very precise, because holding the caliper perpendicularly to the surface was difficult. Determining the geometry of the ramp more accurately could help to reduce the discrepancies between the theoretical and experimental results.

6. Conclusion

All in all, the experiment successfully demonstrated the principles of rolling motion and energy conservation, while also highlighting the importance of considering real-world factors such as friction and measurement accuracy in experimental physics. It was a lot of fun to do this experiment, and I learned a lot about the physics of rolling motion and the importance of careful measurement and analysis in experimental physics. I also learned how to use the precise light barrier system and the stopwatch system, which was very interesting.

List of Figures

4.1. Schematic representation of the inclined plane with height h and length L . [1]	10
4.2. Comparison of theoretical and experimental accelerations for solid and hollow cylinders.	11
4.3. Schematic representation of the height difference h and the length of the ramp L	12

List of Tables

4.1. Parameters of the cylinders used in the experiment.	9
4.2. Measured times and calculated speeds for the solid and hollow cylinders.	13
4.3. Calculated kinetic energies for the solid and hollow cylinders.	13

References

- [1] Dr. J.Wagner, *Physikalisches Anfängerpraktikum*, Stand 11/2024
- [2] W. Ilberg, M. Krötzsch, *Physikalisches Praktikum für Anfänger*, BSB B. G. Teubner Verlagsgesellschaft, Leipzig (1985).

A. Code

height of track

```
[81]: import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
```

```
[82]: h = np.array([11.1, 12.8, 11.5, 12.8, 12.7])
err_h = 0.05

print(np.mean(h))
error_mean = np.std(h, ddof=1)/np.sqrt(len(h))
print(error_mean)
print(np.sqrt(error_mean**2 + err_h**2))
```

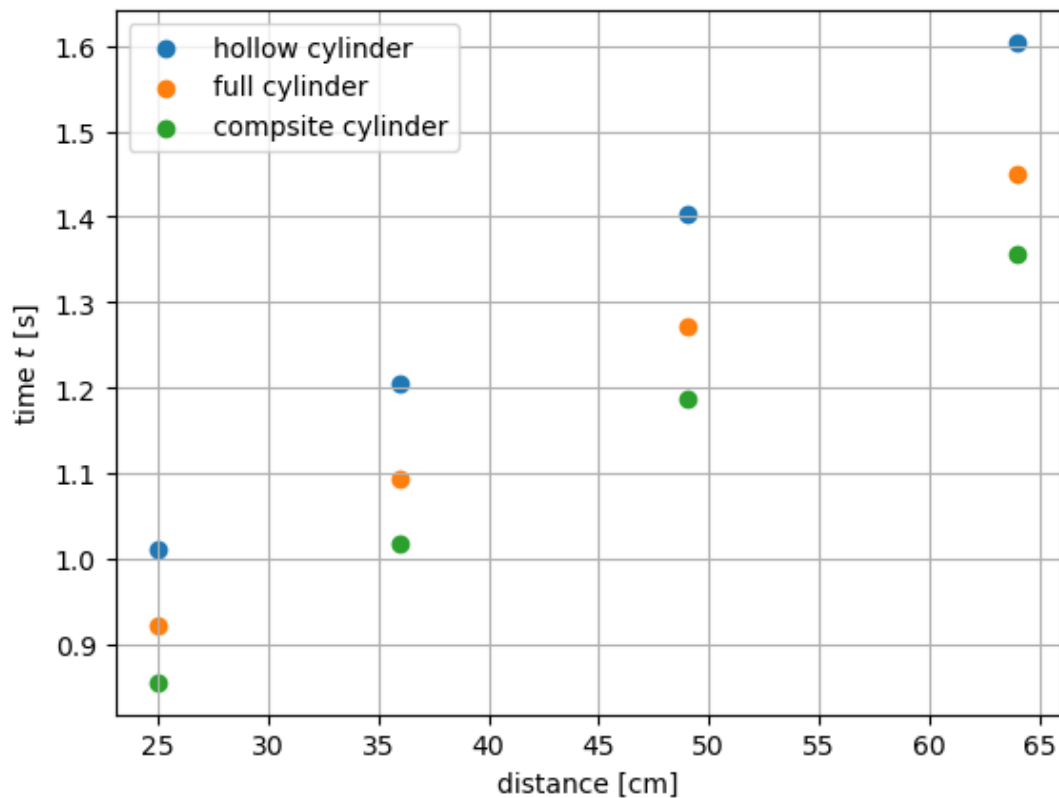
```
12.180000000000001
0.3652396473549936
0.3686461718233353
```

```
[83]: t_hollow = np.array([1.011, 1.204, 1.403, 1.604])
t_full = np.array([0.921, 1.094, 1.272, 1.451])
t_compose = np.array([0.855, 1.018, 1.186, 1.356])

x = np.array([25, 36, 49, 64])

plt.scatter(x, t_hollow, label="hollow cylinder")
plt.scatter(x, t_full, label="full cylinder")
plt.scatter(x, t_compose, label="compsite cylinder")
plt.xlabel("distance [cm]")
plt.ylabel("time $t$ [s]")
plt.grid(True)
plt.legend()
```

```
[83]: <matplotlib.legend.Legend at 0x1520a7d90>
```



Quantitative A3:

```
[84]: x = np.array([25, 36, 49, 64])

t_hollow = np.array([
    [1.019, 1.212, 1.410, 1.610],
    [1.027, 1.223, 1.426, 1.629],
    [1.014, 1.209, 1.410, 1.613],
    [1.022, 1.215, 1.415, 1.614],
    [1.009, 1.201, 1.400, 1.600]
])
t_full = np.array([
    [0.922, 1.096, 1.278, 1.459],
    [0.893, 1.068, 1.245, 1.425],
    [0.900, 1.075, 1.255, 1.435],
    [0.901, 1.074, 1.254, 1.435],
    [0.913, 1.087, 1.266, 1.446]
])

mean_hollow = np.mean(t_hollow, axis=0)
std_hollow = np.std(t_hollow, axis=0, ddof=1)/np.sqrt(t_hollow.
    ↪shape[0]) # Standard error of the mean
mean_full = np.mean(t_full, axis=0)
```

```
std_full = np.std(t_full, axis=0, ddof=1)/np.sqrt(t_full.shape[0]) #  
    ↳Standard error of the mean  
  
print("Hollow cylinder:")  
for distance, mean, std in zip(x, mean_hollow, std_hollow):  
    print(f"{distance} cm: mean = {mean:.4f} s, std = {std:.4f} s")  
  
print("\nFull cylinder:")  
for distance, mean, std in zip(x, mean_full, std_full):  
    print(f"{distance} cm: mean = {mean:.4f} s, std = {std:.4f} s")
```

Hollow cylinder:

```
25 cm: mean = 1.0182 s, std = 0.0031 s  
36 cm: mean = 1.2120 s, std = 0.0036 s  
49 cm: mean = 1.4122 s, std = 0.0042 s  
64 cm: mean = 1.6132 s, std = 0.0047 s
```

Full cylinder:

```
25 cm: mean = 0.9058 s, std = 0.0052 s  
36 cm: mean = 1.0800 s, std = 0.0050 s  
49 cm: mean = 1.2596 s, std = 0.0057 s  
64 cm: mean = 1.4400 s, std = 0.0058 s
```

a4

```
[85]: # messung, t1, t2  
  
# hollow  
m1h = np.array([1.922, 2.192])  
m2h = np.array([1.951, 2.222])  
m3h = np.array([1.936, 2.206])  
m4h = np.array([1.922, 2.192])  
m5h = np.array([1.938, 2.207])  
  
# full  
m1f = np.array([1.724, 1.967])  
m2f = np.array([1.736, 1.980])  
m3f = np.array([1.718, 1.961])  
m4f = np.array([1.737, 1.981])  
m5f = np.array([1.749, 1.992])  
  
m_diff_h = np.array([])  
for i in [m1h, m2h, m3h, m4h, m5h]:  
    m_diff_h = np.append(m_diff_h, i[1] - i[0])  
  
m_diff_f = np.array([])  
for i in [m1f, m2f, m3f, m4f, m5f]:  
    m_diff_f = np.append(m_diff_f, i[1] - i[0])
```

```

err_t = 0.001
print("Hollow cylinder time differences:")
print(m_diff_h)
print("Mean time difference for hollow cylinder:", np.mean(m_diff_h))
print("Standard error for hollow cylinder:", np.std(m_diff_h, ddof=1)/
      ↪len(m_diff_h))
print('Full error for hollow cylinder:', np.sqrt((np.std(m_diff_h,
      ↪ddof=1)/len(m_diff_h))**2 + err_t**2))

print("\nFull cylinder time differences:")

print(m_diff_f)
print("Mean time difference for full cylinder:", np.mean(m_diff_f))
print("Standard error for full cylinder:", np.std(m_diff_f, ddof=1)/
      ↪len(m_diff_f))
print('Full error for full cylinder:', np.sqrt((np.std(m_diff_f,
      ↪ddof=1)/len(m_diff_f))**2 + err_t**2))

```

Hollow cylinder time differences:

[0.27 0.271 0.27 0.27 0.269]

Mean time difference for hollow cylinder: 0.27000000000000001

Standard error for hollow cylinder: 0.00014142135623730964

Full error for hollow cylinder: 0.0010099504938362077

Full cylinder time differences:

[0.243 0.244 0.243 0.244 0.243]

Mean time difference for full cylinder: 0.2434

Standard error for full cylinder: 0.00010954451150102927

Full error for full cylinder: 0.001005982107196743

Examination

```

[86]: import uncertainties as unc
      import uncertainties.unumpy as unp

```

```

[87]: m = unc.ufloat(443, 1) * 1e-3      # mass in kg
      d_outer = unc.ufloat(50, 0.05) * 1e-3 # outer diameter in m
      d_inner = unc.ufloat(46.5, 0.05) * 1e-3 # inner diameter in m

      I_hollow = (1/2) * m * ((d_outer/2)**2 + (d_inner/2)**2)
      I_full = (1/2) * m * (d_outer/2)**2

      print(f"Hollow cylinder moment of inertia: {I_hollow:.7e} kgm2")
      print(f"Solid cylinder moment of inertia: {I_full:.7e} kgm2")

      h = unc.ufloat(10.1, 0.4)*1e-2 # height in m
      l = unc.ufloat(1.000, 0.005) # length in m\

      varphi = unp.arcsin(h / l)

```

```
varphi_deg = varphi * 180 / np.pi
print(f"Angle of inclination: {varphi_deg:.3f} degrees, {varphi:.4f}␣
↪radians")
```

Hollow cylinder moment of inertia: $(2.5817209 \pm 0.0069469) \times 10^{-4} \text{ kgm}^2$

Solid cylinder moment of inertia: $(1.3843750 \pm 0.0041751) \times 10^{-4} \text{ kgm}^2$

Angle of inclination: 5.797 ± 0.232 degrees, 0.1012 ± 0.0041 radians

```
[88]: a_solid = (9.81 * unp.sin(varphi)) / (1 + I_full / (m * (d_outer/2)**2))
a_hollow = (9.81 * unp.sin(varphi)) / (1 + I_hollow / (m * (d_outer/
↪2)**2))

print(a_solid)
print(a_hollow)
```

0.661 ± 0.026

0.513 ± 0.020

Plot the values in Diagram

```
[89]: mean_hollow_squared = mean_hollow**2
std_hollow_squared = np.sqrt(std_hollow**2 + 0.001**2) * mean_hollow * 2
mean_full_squared = mean_full**2
std_full_squared = np.sqrt(std_full**2 + 0.001**2) * mean_full * 2

s = np.array([25, 36, 49, 64])*1e-2

plt.errorbar(s, mean_full_squared, yerr=std_full_squared, fmt='.',␣
↪label="Full Cylinder", capsize=3, color='orange')

plt.errorbar(s, mean_hollow_squared, yerr=std_hollow_squared, fmt='.',␣
↪label="Hollow Cylinder", capsize=3, color='blue')

func = lambda x, a: (2.0 / a) * x

popt_hollow, pcov_hollow = curve_fit(func, s, mean_hollow_squared,␣
↪sigma=std_hollow_squared, absolute_sigma=True)
popt_full, pcov_full = curve_fit(func, s, mean_full_squared,␣
↪sigma=std_full_squared, absolute_sigma=True)

x_fit = np.linspace(0, 1, 100)
plt.plot(x_fit, func(x_fit, popt_full[0]), label=f"Fit Full Cylinder $a␣
↪= {popt_full[0]:.4f}\, \text{m/s}^2$", color='red')
```

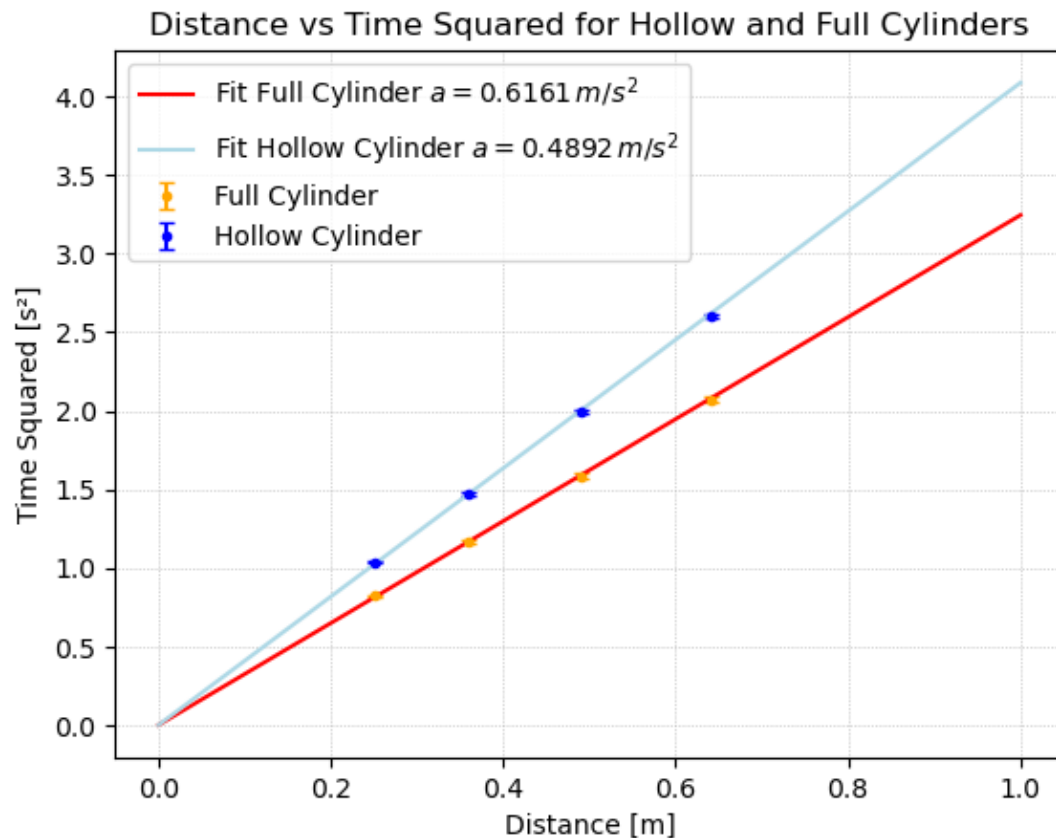
```
plt.plot(x_fit, func(x_fit, popt_hollow[0]), label=f"Fit Hollow
↳Cylinder $a = {popt_hollow[0]:.4f}\, m/s^2$", color='lightblue')

plt.xlabel("Distance [m]")
plt.ylabel("Time Squared [s²]")
plt.grid(which='both', linestyle=':', linewidth=0.5)
plt.legend()
plt.title("Distance vs Time Squared for Hollow and Full Cylinders")
print(f"Full cylinder fit parameters: slope = {popt_full[0]:.4f}
↳$\pm$ {np.sqrt(pcov_full[0][0]):.4f}")
print(f"Hollow cylinder fit parameters: slope = {popt_hollow[0]:.4f}
↳$\pm$ {np.sqrt(pcov_hollow[0][0]):.4f}")
```

Full cylinder fit parameters: slope = 0.6161 $\pm$ 0.0029

Hollow cylinder fit parameters: slope = 0.4892 $\pm$ 0.0015

```
<>:22: SyntaxWarning: invalid escape sequence '\,'
<>:23: SyntaxWarning: invalid escape sequence '\,'
<>:22: SyntaxWarning: invalid escape sequence '\,'
<>:23: SyntaxWarning: invalid escape sequence '\,'
/var/folders/9r/ppvq7f7j3kv69gnk_v549lcm0000gn/T/ipykernel_18719/
↳668146309.py:22
: SyntaxWarning: invalid escape sequence '\,'
  plt.plot(x_fit, func(x_fit, popt_full[0]), label=f"Fit Full Cylinder
↳$a =
{popt_full[0]:.4f}\, m/s^2$", color='red')
/var/folders/9r/ppvq7f7j3kv69gnk_v549lcm0000gn/T/ipykernel_18719/
↳668146309.py:23
: SyntaxWarning: invalid escape sequence '\,'
  plt.plot(x_fit, func(x_fit, popt_hollow[0]), label=f"Fit Hollow
↳Cylinder $a =
{popt_hollow[0]:.4f}\, m/s^2$", color='lightblue')
```



Fitted the other way around

```
[93]: mean_hollow_squared = mean_hollow**2
std_hollow_squared = np.sqrt(std_hollow**2 + 0.001**2) * mean_hollow * 2
mean_full_squared = mean_full**2
std_full_squared = np.sqrt(std_full**2 + 0.001**2) * mean_full * 2

s = np.array([25, 36, 49, 64]) * 1e-2

plt.errorbar(mean_full_squared, s, xerr=std_full_squared, fmt='.',
             label='Full Cylinder', capsize=3, color='orange')
plt.errorbar(mean_hollow_squared, s, xerr=std_hollow_squared, fmt='.',
             label='Hollow Cylinder', capsize=3, color='blue')

# keep the same model form y = a*x + c, but fix c = 0
func = lambda x, a, c=0: (a / 2.0) * x + c

popt_hollow, pcov_hollow = curve_fit(func, mean_hollow_squared, s,
                                     sigma=np.full_like(s, 0.005), absolute_sigma=True, p0=[1.0],
                                     bounds=([-np.inf], [np.inf]))
popt_full, pcov_full = curve_fit(func, mean_full_squared, s, sigma=np.
                                 full_like(s, 0.005), absolute_sigma=True, p0=[1.0], bounds=([-np.
                                 inf], [np.inf]))
```

```

x_fit = np.linspace(0, np.max(mean_full_squared) * 1.5, 100)
plt.plot(x_fit, func(x_fit, *popt_full), label=f'Fit Full Cylinder $a = \{popt\_full[0]:.4f\} \backslash, m/s^2$', color='red')
plt.plot(x_fit, func(x_fit, *popt_hollow), label=f'Fit Hollow Cylinder $a = \{popt\_hollow[0]:.4f\} \backslash, m/s^2$', color='lightblue')

plt.xlabel('Time Squared [s^2]')
plt.ylabel('Distance [m]')
plt.grid(which='both', linestyle=':', linewidth=0.5)
plt.legend()
plt.title('Time Squared vs Distance for Hollow and Full Cylinders')
print(f'Full cylinder fit parameters: a = {popt_full[0]:.4f} $\pm$ {np.sqrt(pcov_full[0][0]):.4f}, c = 0')
print(f'Hollow cylinder fit parameters: a = {popt_hollow[0]:.4f} $\pm$ {np.sqrt(pcov_hollow[0][0]):.4f}, c = 0')

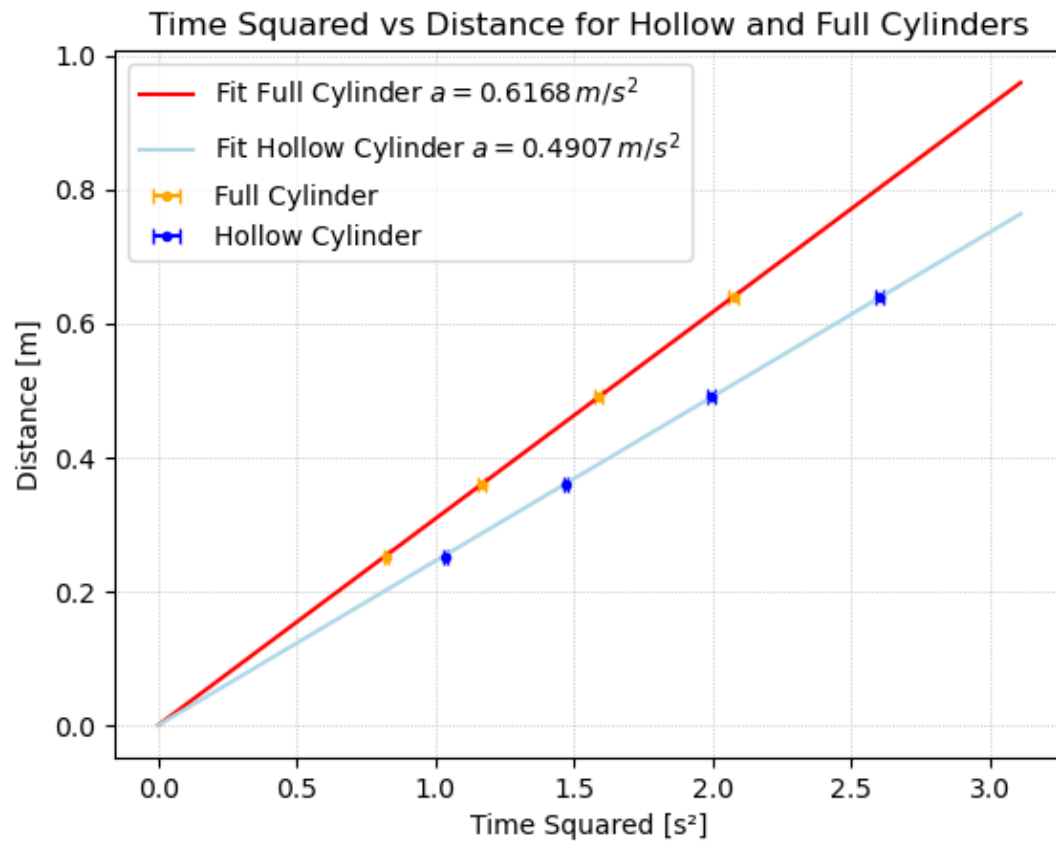
```

```

<>:18: SyntaxWarning: invalid escape sequence '\,'
<>:19: SyntaxWarning: invalid escape sequence '\,'
<>:18: SyntaxWarning: invalid escape sequence '\,'
<>:19: SyntaxWarning: invalid escape sequence '\,'
/var/folders/9r/ppvq7f7j3kv69gnk_v549lcm0000gn/T/ipykernel_18719/
↳3647935948.py:1
8: SyntaxWarning: invalid escape sequence '\,'
    plt.plot(x_fit, func(x_fit, *popt_full), label=f'Fit Full Cylinder $a = \{popt\_full[0]:.4f\} \backslash, m/s^2$', color='red')
/var/folders/9r/ppvq7f7j3kv69gnk_v549lcm0000gn/T/ipykernel_18719/
↳3647935948.py:1
9: SyntaxWarning: invalid escape sequence '\,'
    plt.plot(x_fit, func(x_fit, *popt_hollow), label=f'Fit Hollow Cylinder $a = \{popt\_hollow[0]:.4f\} \backslash, m/s^2$', color='lightblue')

Full cylinder fit parameters: a = 0.6168 $\pm$ 0.0034, c = 0
Hollow cylinder fit parameters: a = 0.4907 $\pm$ 0.0027, c = 0

```



Calculate Height

```
[ ]: d = unc.ufloat(11, 0.5)*1e-3 # dist start in m

h = h/l * (1-d)
print('delta h in m:', h)
```

delta h in m: 0.100+/-0.004

```
[ ]: E = m * 9.81 * h
E
```

```
[ ]: 0.43410061287+/-0.01722142256704715
```

```
[ ]: m_diff_f
err_f = np.sqrt((np.std(m_diff_f, ddof=1)/len(m_diff_f))**2 + err_t**2)

m_diff_h
err_h = np.sqrt((np.std(m_diff_h, ddof=1)/len(m_diff_h))**2 + err_t**2)

d = unc.ufloat(25, 0.07)*1e-2 # dist between light barriers in m

t_f = unc.ufloat(np.mean(m_diff_f), err_f)
```

```

t_h = unc.ufloat(np.mean(m_diff_h), err_h)

v_f = d / t_f
v_h = d / t_h

r = d_outer / 2

print(f"Full cylinder speed: {v_f:.4f} m/s")
print(f"Hollow cylinder speed: {v_h:.4f} m/s")

w_f = v_f / r
print(f"Full cylinder angular velocity: {w_f:.4f} rad/s")
w_h = v_h / r
print(f"Hollow cylinder angular velocity: {w_h:.4f} rad/s")

E_trans_f = 1/2 * m * v_f**2
E_rot_f = 1/2 * I_full * w_f**2

E_trans_h = 1/2 * m * v_h**2
E_rot_h = 1/2 * I_hollow * w_h**2

print(f"\nFull cylinder translational kinetic energy: {E_trans_f:.5f} J")
print(f"Full cylinder rotational kinetic energy: {E_rot_f:.5f} J")
print(f"total kinetic energy: {E_trans_f + E_rot_f:.5f} J")
print(f"\nHollow cylinder translational kinetic energy: {E_trans_h:.5f} J")
print(f"Hollow cylinder rotational kinetic energy: {E_rot_h:.5f} J")
print(f"total kinetic energy: {E_trans_h + E_rot_h:.5f} J")

```

Full cylinder speed: 1.0271+/-0.0051 m/s
Hollow cylinder speed: 0.9259+/-0.0043 m/s
Full cylinder angular velocity: 41.0846+/-0.2092 rad/s
Hollow cylinder angular velocity: 37.0370+/-0.1770 rad/s

Full cylinder translational kinetic energy: 0.23368+/-0.00239 J
Full cylinder rotational kinetic energy: 0.11684+/-0.00120 J
total kinetic energy: 0.35051+/-0.00359 J

Hollow cylinder translational kinetic energy: 0.18990+/-0.00183 J
Hollow cylinder rotational kinetic energy: 0.17707+/-0.00172 J
total kinetic energy: 0.36697+/-0.00354 J

[]: